

A GEOSPATIAL ANALYSIS OF THE SUNSPOTS EFFECT ON GROUNDWATER LEVELS IN LONG ISLAND, NY DURING 2002-2021

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Outline

- Decomposition of time series.
- Correlation between the long-term component of the groundwater and the daily sunspot number.
- Correlation between the long term of the daily change of sunspot number and groundwater.
- Conclusion

Introduction

- We study the long-term relationship between the groundwater and the daily sunspot number in several stations in Long Island, New York.
- For the analysis, we use the time series decomposition, and we analyze the data based on the main periodicities of the groundwater and the sunspot data.

Introduction

- Environmental and climatic data consists of different signals which they need to be separated to reveal the “real” peaks of the studied variables.
- The presence of the short term variations or noise in the time series data can change the inferences of the analysis.
- Aim: Removing the sunspots number effect may facilitate a better understanding of another factor such as a human factor on the groundwater resources

Data

- We use daily groundwater and daily sunspot number data measured from 2002 to 2021.
- Sunspots Number derived from R software and also from (<http://sidc.oma.be/silso/INFO/sndtotcsv.php>)
- The data have been derived from 61 USGS groundwater monitoring stations in Long Island, New York.

Methodology

- Application of the Time Series Decomposition in groundwater and sunspot numbers.
- Application of the KZ filter to remove the short-term variations from the time series.
- Correlation between the long-term component of the groundwater and the daily sunspot number.
- Correlation between the long term of the daily absolute change of sunspot number and groundwater.

Kolmogorov-Zurbenko Filter ($KZ_{m,p}$)

- For the decomposition of the time series we use the Kolmogorov-Zurbenko filter. The filter is defined by:

$$Y_t = \frac{1}{m} \sum_{j=-k}^k X_{t+j}$$

$$m=2k+1$$

X_t : original series

Y_t : filtered series

p : number of iterations

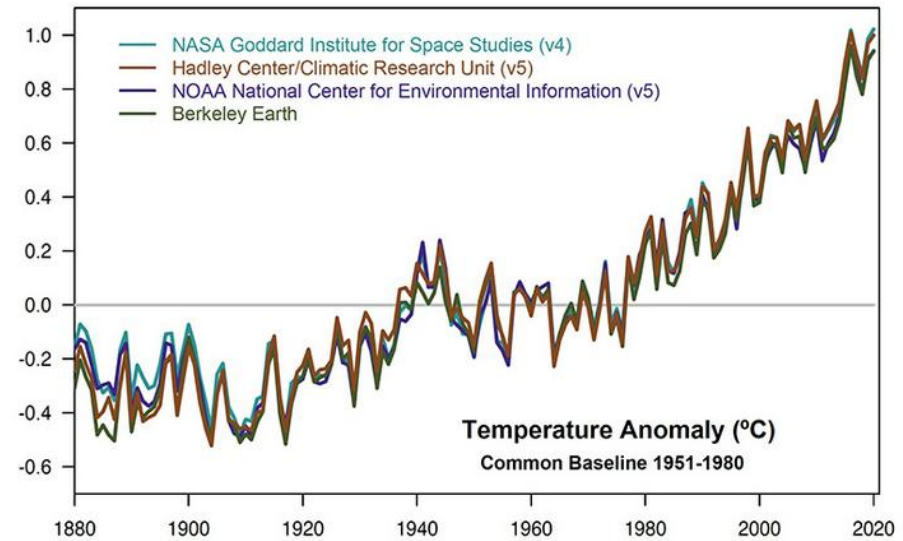
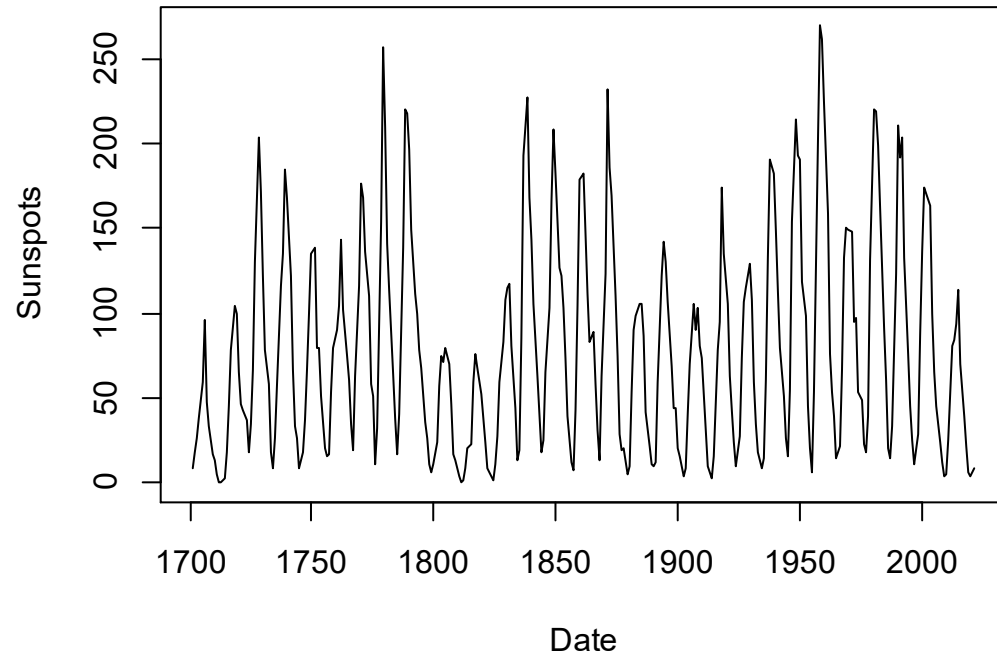
Decomposition of time series

$$X(t) = L(t) + Se(t) + Sh(t)$$

- $X(t)$: original time series
- $L(t)$: long term trend component
- $Se(t)$: seasonal component
- $Sh(t)$: short term component

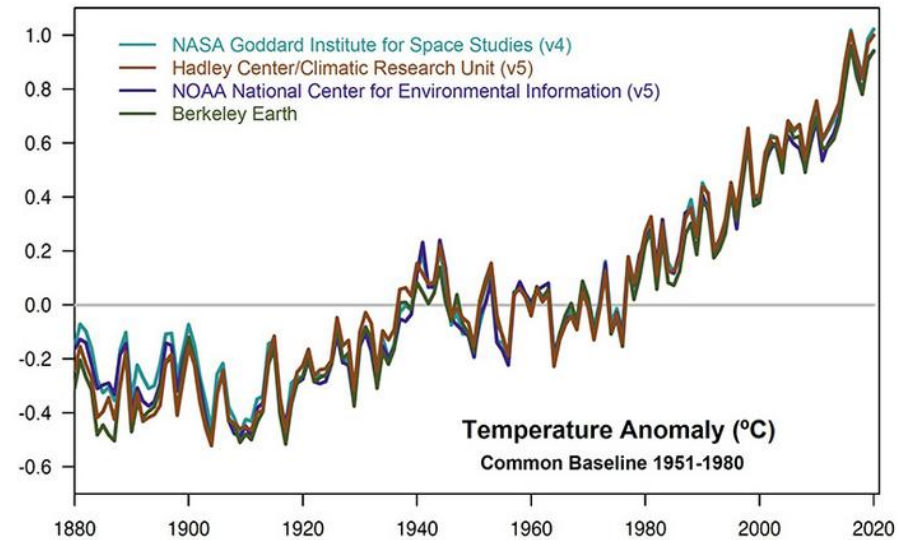
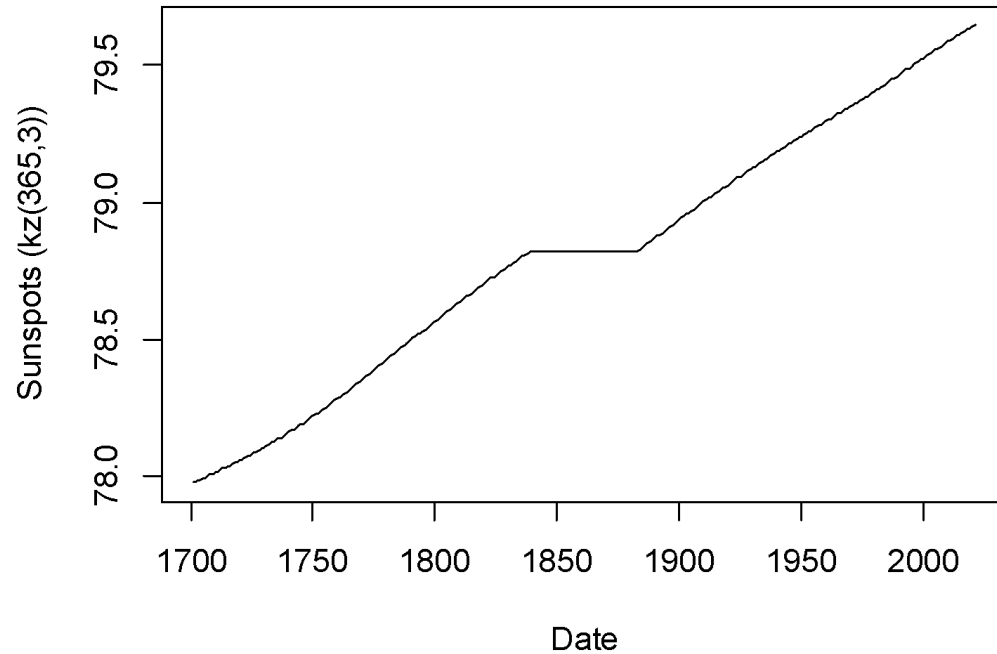
Annual data of Sunspots compared with climate change from 1700 to 2021

ANNUAL DATA



Long term component of Sunspots compared with climate change from 1700 to 2021

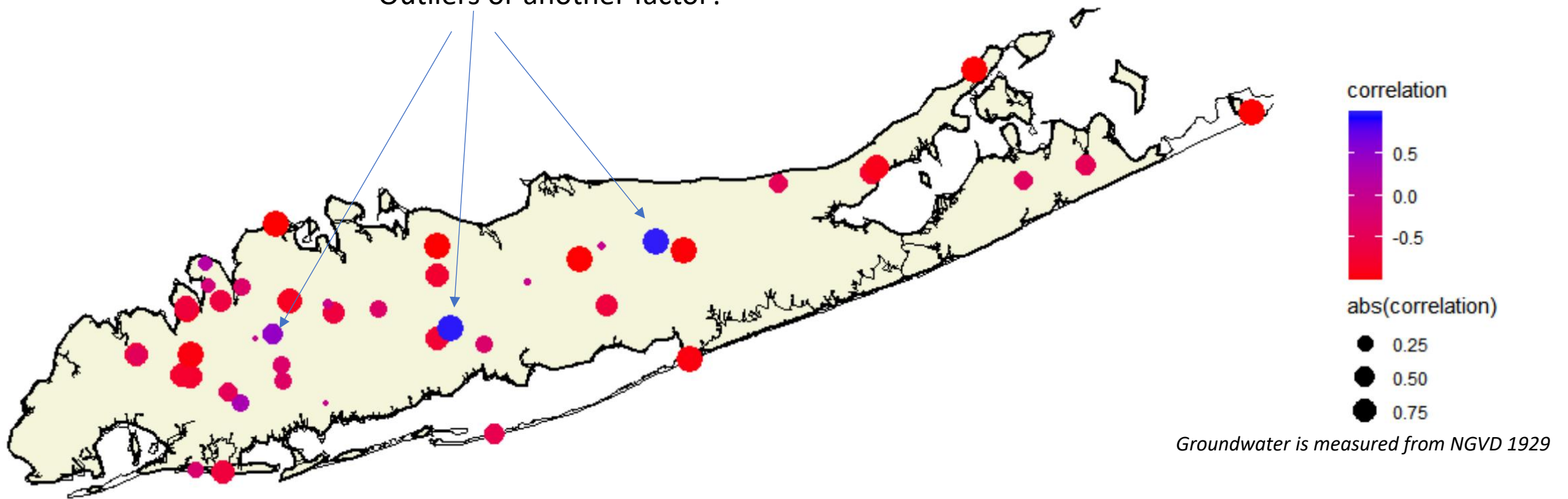
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Negative correlation between Groundwater versus Sunspot Numbers in Long Island, New York

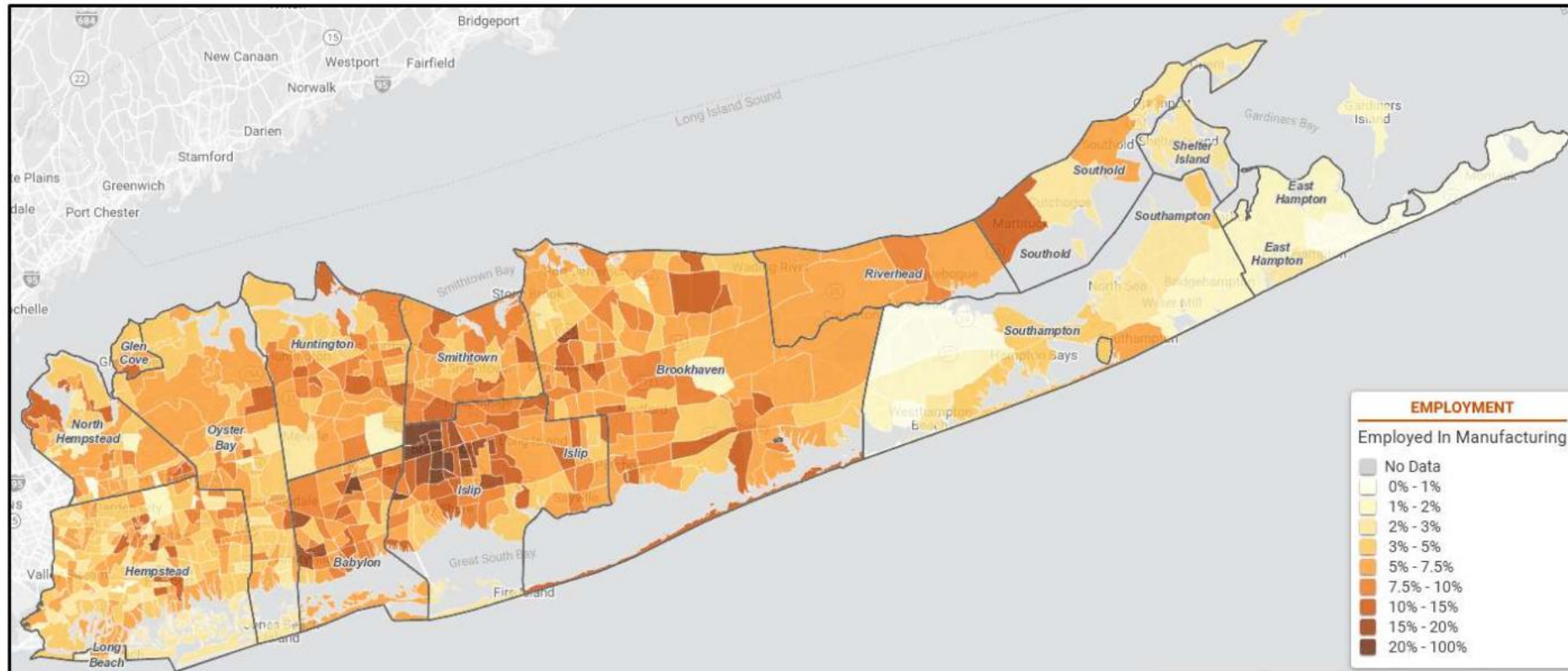
Long Island, NY

Outliers or another factor?



Negative correlation between Groundwater versus Sunspot Numbers in Long Island, New York

Long Island Index provided Demographic maps (Long Island's Changing Demographics)

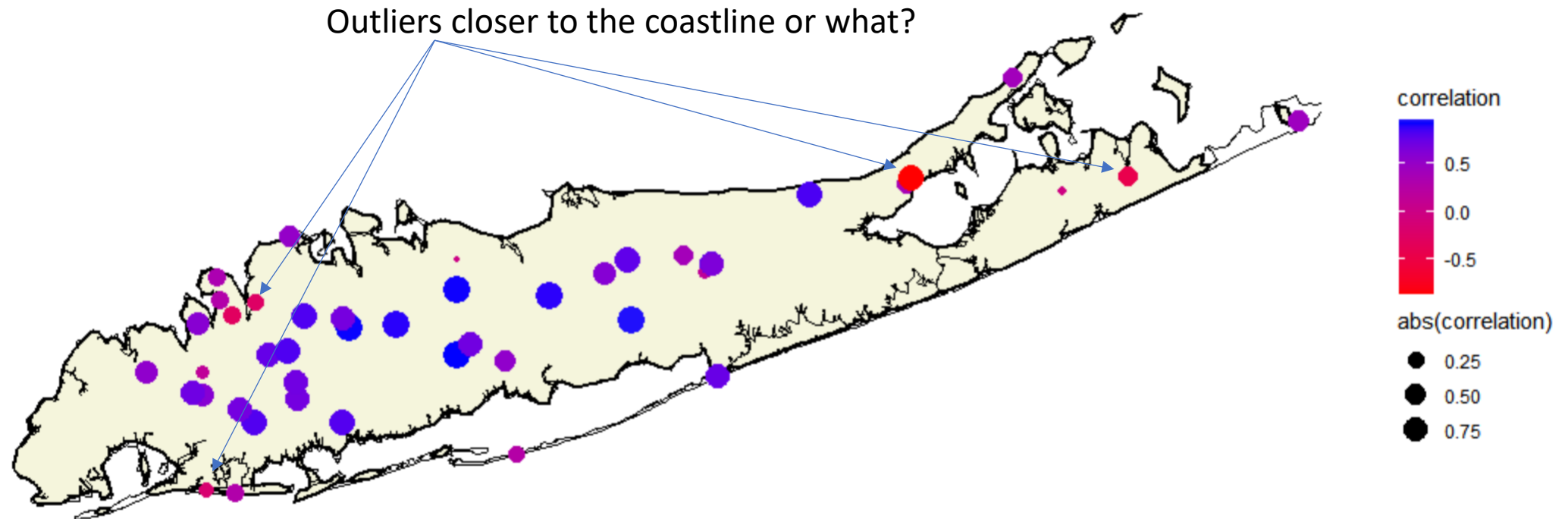


Long Island's demographic map showing employed Long Islanders in manufacturing jobs for the period of 1970-2010.

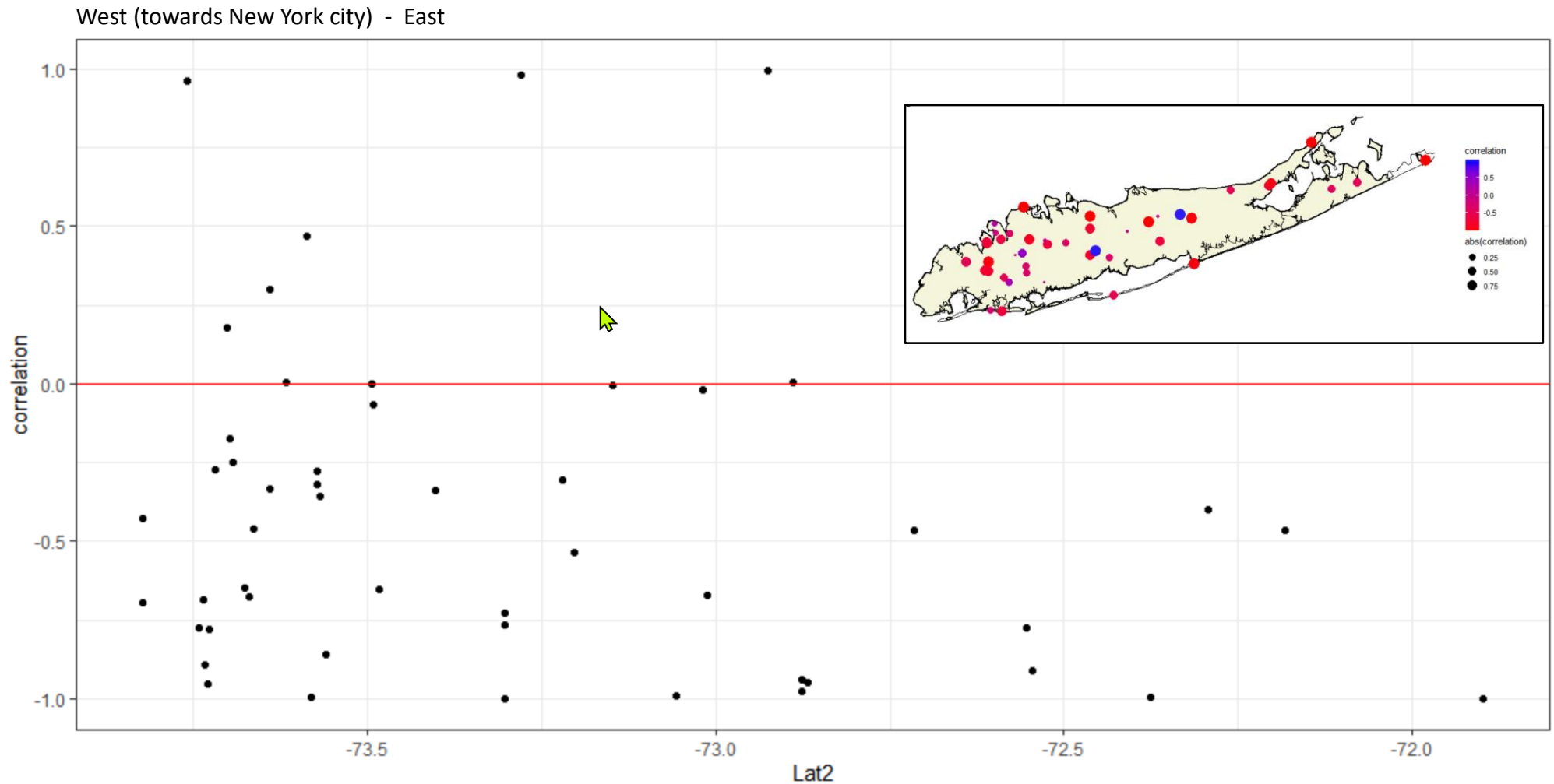
<http://historiccensus.longislandindexmaps.org/?latlng=40.867798%2C72.858149&z=10&variable=PctMnfg&year=2010>

Positive correlation between absolute daily change of Groundwater versus Sunspot Numbers in Long Island, New York

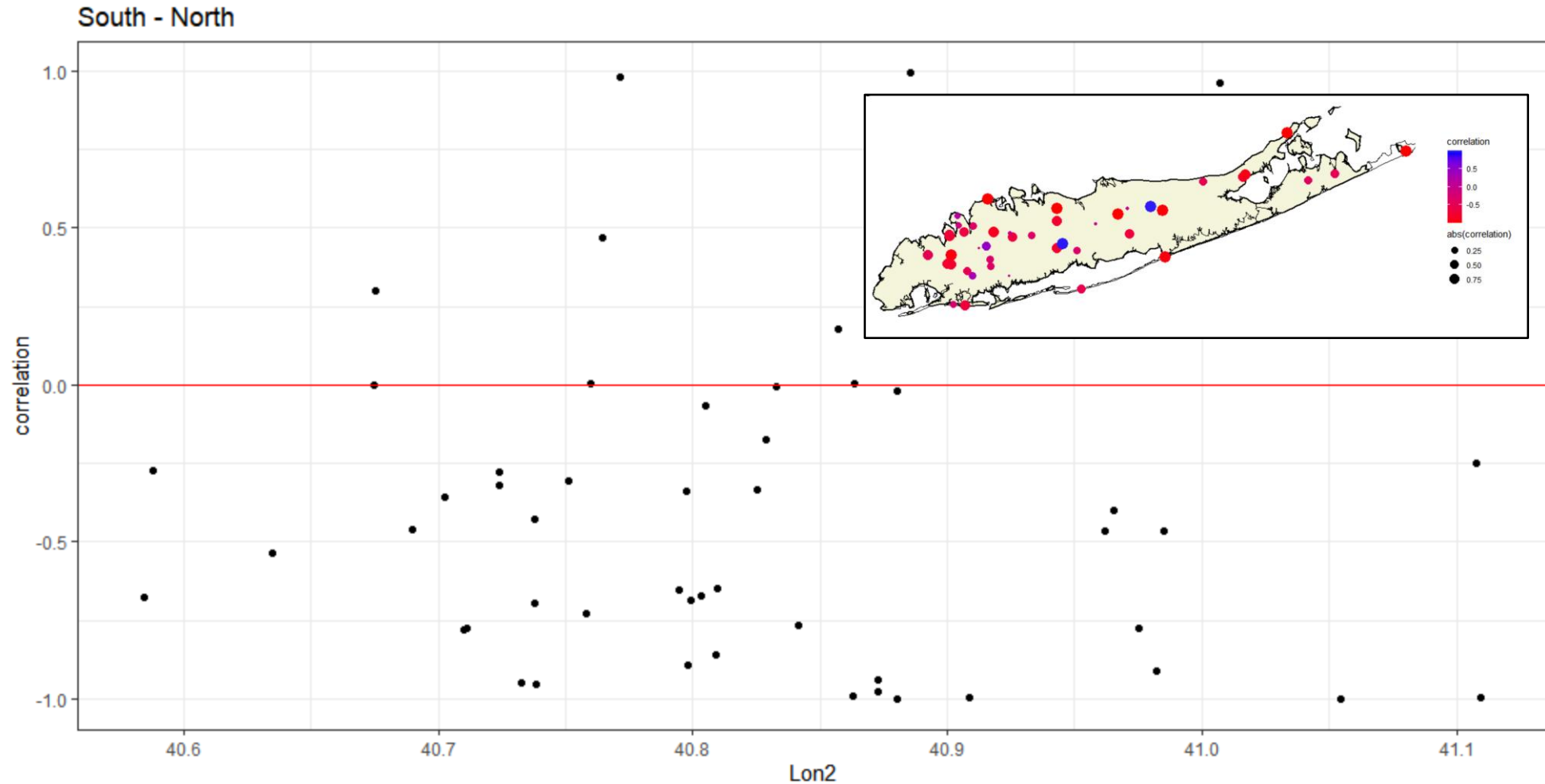
Long Island, NY



Correlation between Groundwater and Sunspot Numbers in Long Island, New York

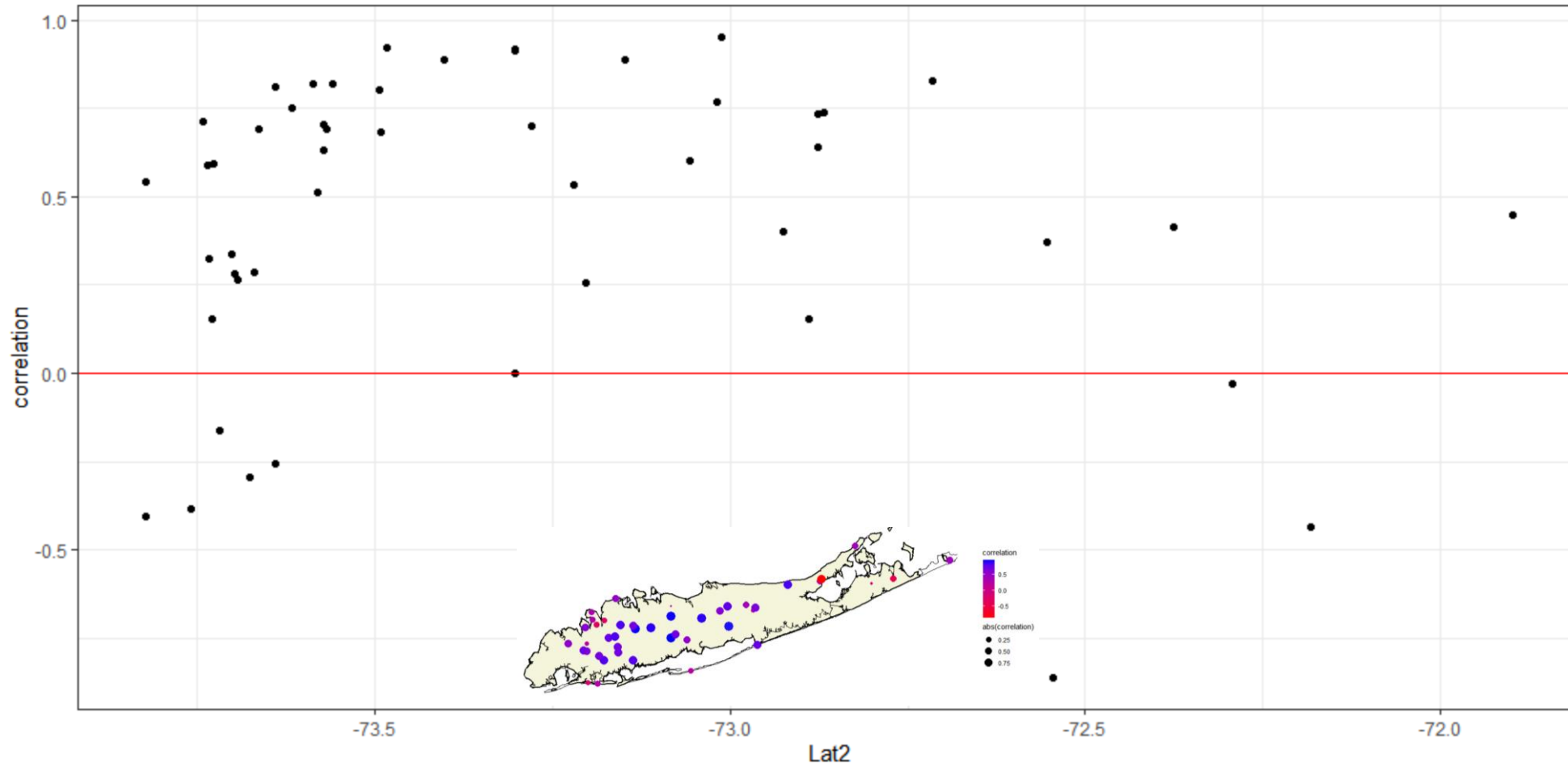


Correlation between Groundwater and Sunspot Numbers in Long Island, New York

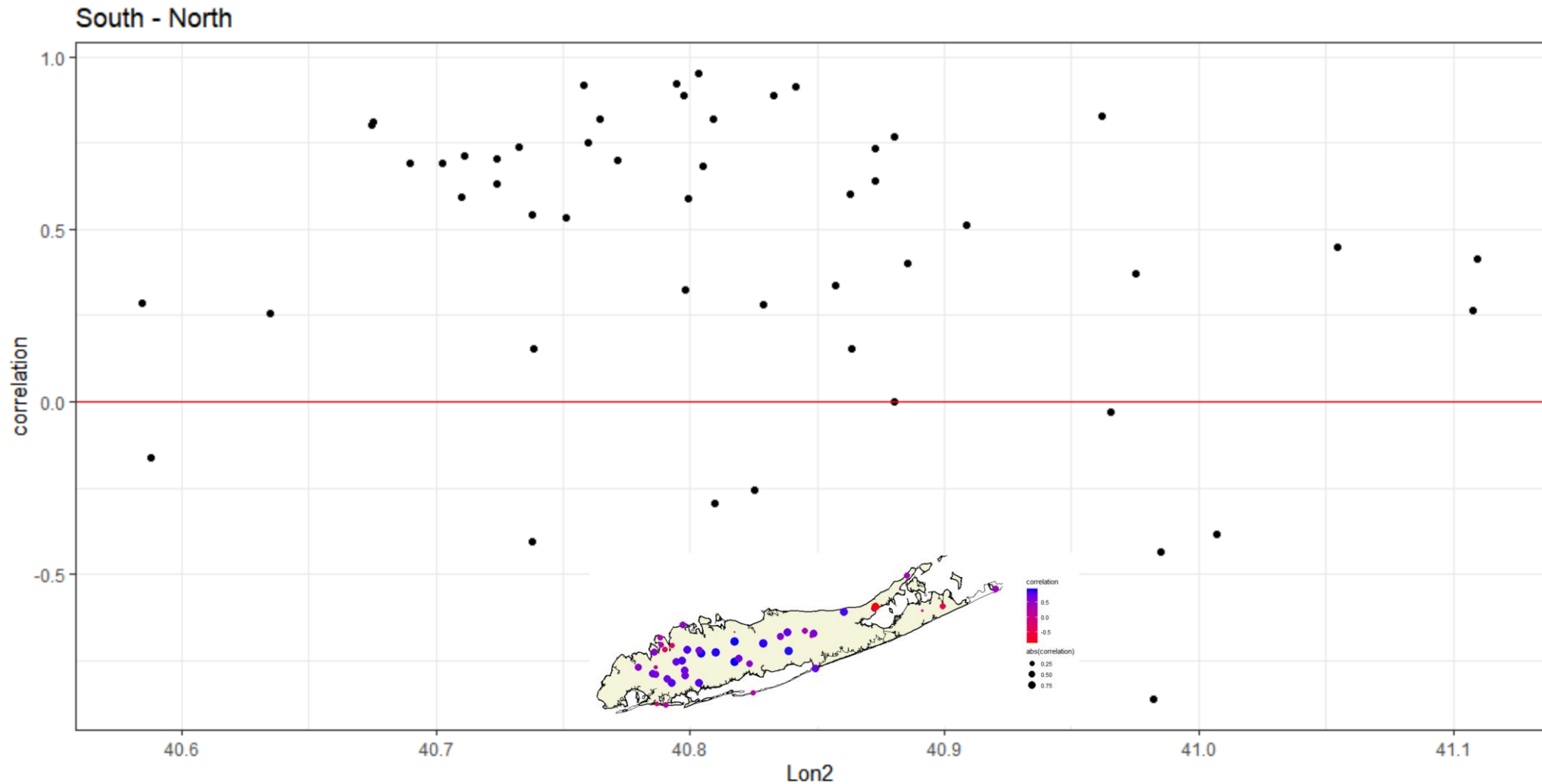


Correlation between Groundwater and Sunspot Numbers absolute change in Long Island, New York

West (towards New York city) - East



Correlation between Groundwater and Sunspot Numbers absolute change in Long Island, New York



Conclusions

- Removing noise and eliminating interferences we may focus on the real factors to water fluctuations
- The long-term association between the groundwater and an exogenic factor (sunspot N) may identify much better endogenic factors such as human activity and new thresholds for hazards.
- Absolute changes of the sunspots number positively correlates with the groundwater's long-term component.
- This correlation can be explained as that sudden sunspot number changes may cause stronger temporal groundwater influences not necessarily related to Earth's system mechanisms.

Thank you.